

Basic Instructions and How To Privacy Analysis of Augmented Reality Glasses

1 Basic Set Up



1. Plug in the ALFA network adapter with a USB adapter.
2. Make sure the laptop is connected to an internet connection.
3. Start the hotspot (I used a bash script provided to me).
3. Connect the phone to the hotspot.
4. Start wireshark with sudo access, if it is not installed:

```
sudo wireshark
```

If wireshark is not installed, write:

```
sudo apt install wireshark
```

5. Choose the interface of the network adapter and start capture.
6. Start experiments by interact with the glasses and the phone to generate traffic.

2 Micro:Bit

Powering on:

Either - Plug the battery pack into the micro bit. A red light should turn on.

Or - Plug in the micro:bit to the laptop using a USB cable. The red light plus another yellow light should turn on.

Flashing a program onto the micro bit:

Plug the micro bit onto the laptop.

One can write code either with MakeCode or on the micro:bit Python Editor.

Download the .hex file (your program) then drag and drop it onto the MICROBIT drive.
An example program that creates a heart can be found in the following link:
<https://microbit.org/get-started/user-guide/reset-your-microbit/>

Re-flash the micro with the original firmware:

Download the .hex file from the link:

<https://microbit.org/get-started/user-guide/firmware/>

The steps are also on the page, but to sum it up:

1. Press and hold on the reset button (same side as the lights, should be on the left side of the yellow light).
2. Plug in the micro bit to the laptop with a USB whilst pressing down on the button.
3. Afterwards, you should see MAINTENANCE, not MICROBIT, on your laptop. Then you can let go of the button.
4. Check one of the three version of the micro bit, this can be found on the bottom right corner (V1, V2.00, V2.20/21).
5. Download the correct version then drag and drop the .hex file onto the drive.
6. Next, wait for the yellow light to stop flashing and when the upgrade is completed, the micro bit will reset.
7. Everything should be done one sees the MICROBIT drive appears on the laptop.

The following errors are what I encountered when using the micro bit:

504 with an unhappy face - indicates that there is an issue during the code upload or execution process. This can be caused if the code is corrupted or if there were issues during file transfer issues.
528 with an unhappy face - indicates a runtime error related to flashing or executing code on the device. This can be caused if the firmware or .hex file flashed was corrupted or incompatible.

3 BLE sniffer

The program that I tried to use and sniff BLE traffic is from

<https://github.com/virtualabs/btlejack>

Btlejack allows sniffing, jamming and hijacking Bluetooth Low Energy devices.

The steps to set up the sniffer:

1. Install the 'btlejack' software with Pip, this would require one to use a virtual environment which can be found below"

```
sudo pip3 install btlejack
```

2. Connect the micro bit device to the laptop with a USB cable.
3. Use the following command which will program every micro bit device connected to the laptop, preparing them for use with Btlejack:

```
btlejack -i
```

4. There should be an output with:

```
Flashing ...  
Flash n device
```

5. Scan and try to get the access address of active BLE devices with:

```
btlejack -s
```

4 Virtual environment

The steps to set up a virtual environment:

1. One will need python3, so if it not installed write:

```
sudo apt install python3
```

2. One will also need the venv module, so if it is not installed write:

```
sudo apt install python3-venv
```

3. Create the virtual environment:

```
python3 -m venv myvenv
```

4. Start the virtual environment:

```
source myvenv/bin/activate
```

5. Then one should see '(myvenv)' in front of the terminal prompt.
6. To stop the virtual environment:

```
deactivate
```

5 Live Stream on Instagram

1. Power on the glasses and make sure they are connected on the Meta View app.
2. Open Instagram.
3. Press the  at the bottom middle.
4. Then swipe left gently till the end where you see 'Live' at the bottom.
5. Make sure to check who do you want seeing the live stream. (Default is everyone)
6. On the right side of the phone, you should see your glasses, press that.
7. Then double press the capture button on your glasses to switch to your glasses camera. Meta should also inform you about this step.
8. Start live stream at the bottom middle circle button.

Note that you can switch between cameras at any time by double pressing the capture button on your glasses.

6 Unpairing the glasses with the phone

Either: In Settings,

1. Go to Bluetooth and unpair the glasses;

Or: In the Meta View app,

1. press settings at the bottom right.
2. Tap 'Your glasses' or directly the device you want to unpair.
3. Tap 'General', 'Unpair glasses', 'Unpair'.

Note that it is easier and simpler to do it in Settings.

7 Re-pairing the glasses to the phone

[https://www.meta.com/en-gb/help/smart-glasses/articles/ray-ban-meta/set-up-ray-ban-meta-smart-gla](https://www.meta.com/en-gb/help/smart-glasses/articles/ray-ban-meta/set-up-ray-ban-meta-smart-glasses)

1. Take the glasses out of the case and power on.
2. Put the glasses back inside the case and close the case.
3. Wait a bit for the LED on the case to start blinking blue, this means that the device is ready to pair with the Meta View app. If the LED does not blink blue, press and hold the button on the back of the case for at least 5 seconds till you see blinking blue.
4. On the Meta View app, press settings at the bottom right and then follow the on-screen instructions to pair the device.

8 Factory Reset the glasses

Link: [https://www.meta.com/en-gb/help/smart-glasses/articles/ray-ban-meta/factory-reset-ray-ban-me](https://www.meta.com/en-gb/help/smart-glasses/articles/ray-ban-meta/factory-reset-ray-ban-meta-smart-glasses)

1. Power on the glasses and only need to close the right arm.
2. Press the capture button 5 times VERY quickly then you should see the notification LED turn orange.

3. Press and hold the capture button until that notificaiton LED starts blinking.
4. Then release the capture button wait for the LED to turn blue that indicates that you can now pair to a new device.